

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a strong foundation in Verilog, Python, and FPGA hardware design. Experienced in implementing innovative solutions and eager to contribute to cutting-edge projects within a dynamic organization that prioritizes technical expertise and continuous learning.

EXPERIENCE

Intern

3D Cubic, Surat

Aug 2021 - Oct 2021

- Completed an internship focused on understanding and operating 3D printing technology, enhancing practical knowledge of additive manufacturing processes.
- Designed and fabricated a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, demonstrating hands-on expertise and attention to detail in production.

EDUCATION

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, FPGA Design, Cadence, Synopsys VCS, Primetime, Quartus II IDE, MATLAB, CAD, Proteus, P-spice, Simplify3D, MobaXterm, TCL, Perl, Basic C, G-Code, PCI-E, PCI, Intel DE-10 Lite FPGA Board, Oscilloscopes, DMM, Signal Generators, Sensors, Probes, Bread Board

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed SRAM architecture with detailed block and state diagrams for PCI to SRAM transactions, ensuring compatibility with 64- and 32-bit bus standards.
- Prepared comprehensive timing diagrams that facilitated accurate data transmission, achieving a 20% reduction in transaction latency.

Vending Machine

Aug 2023 - Dec 2023

- Integrated finite state machines for idle, selection, payment, dispensing, and reset states, resulting in a seamless user experience.
- Tested and verified the design using Verilog testbenches and simulation environments, achieving a 95% success rate in hardware validation on an Intel DE-10 Lite FPGA board.

6-Bit Segmented DAC

Jan 2024 - May 2024

- Divided DAC implementation into two segments: LSB and MSB, achieving a 15% improvement in output stability and accuracy.
- Successfully added two bits to the design, resulting in expected values of 12V, meeting performance benchmarks.

STA for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints scripts to generate comprehensive setup and hold timing reports, identifying and correcting timing violations that improved design performance by 30%.
- Optimized generated slack through targeted adjustments, enhancing overall system reliability and efficiency.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented a Verilog module and used a Perl script to verify functionality, achieving a 100% accuracy rate in output results.

- Generated detailed timing reports by applying design constraints, ensuring compliance with static timing analysis standards.